

5. FET CHARACTERISTICS

AIM:

a) To draw the drain and transfer characteristics of a given FET.

APPARATUS:

FET BFW11	-1No.
Regulated power supply (0-30V)	-1No.
Voltmeter (0-20V)	-2No.
Ammeter (0-20mA)	-1No.
Bread board	
Connecting wires	

THEORY:

A FET is a three-terminal device, in which current conduction is by majority carriers only. The flow of current is controlled by means of an Electric field. The three terminals of FET are Gate, Drain and Source. It is having the characteristics of high input impedance and less noise, the Gate to Source junction of the FETs always reverse biased. In response to small applied voltage from drain to source, the n-type bar acts as sample resistor, and the drain current increases linearly with V_{DS} . With increase in I_D the ohmic voltage drop between the source and the channel region reverse biases the junction and the conducting position of the channel begins to remain constant. The V_{DS} at this instant is called “pinch of voltage”. If the gate to source voltage (V_{GS}) is applied in the direction to provide additional reverse bias, the pinch off voltage will be decreased. In amplifier application, the FET is always used in the region beyond the pinch-off.

FET parameters:

AC Drain Resistance, $r_d = \Delta V_{DS} / \Delta I_D$ at constant V_{GS}

Trans conductance, $g_m = \Delta I_D / \Delta V_{GS}$ at constant V_{DS}

Amplification, $\mu = \Delta V_{DS} / \Delta V_{GS}$ at constant I_D

Relation between above parameters

$$\mu = r_d * g_m$$

The drain current is given by

CEC